Adhyan Ahuja

Data Scientist

Hyderabad, India | adhyanahuja17@gmail.com | +91-9881401378 | linkedin.com/in/adhyanahuja10

Professional Summary

Data Scientist with 2 years of experience in Analytics, Applied ML, and GenAI-driven solutions. Currently working at D. E. Shaw, supporting NY head-office stakeholders through scalable data products, automated reporting, and insight generation. Strong foundation in Data Science and Mathematics with specialized focus on ML and Statistical Modeling.

Education

Birla Institute of Technology and Science, Pilani — Aug 2019 – Jun 2024

B.E. in Electrical and Electronics Engineering, M.Sc. in Mathematics; Minor in Data Science — **GPA: 8.44/10**

- **Key Achievements:** Led 10+ events as Mathematics Association coordinator, boosting participation through data-driven strategies; Campus Ambassador for AAROHAN IQ exam, successfully attracting 1,200+ participants.
- **Relevant Coursework:** Foundations of Data Science, Machine Learning, Deep Learning, Applied Statistical Methods

Delhi Public School, Pune — Graduated: May 2019

Class 10: CGPA 10.0/10, Class 12: 94.2%

Professional Experience

Specialist - Global Analytics & Reporting | D. E. Shaw India Private Limited | Hyderabad, India — Dec 2025 – Present

- **Title Recommendation Engine:** Built a hybrid rule-based and LLM-powered system to recommend officer titles for incoming candidates, combining structured ranking heuristics with LLM-driven qualitative evaluation to improve title alignment and reduce manual review effort.
- **Analytics Dashboards Development:** Built interactive dashboards to track key hiring and learning metrics, enabling stakeholders to self-serve insights and reducing dependency on manual data pulls.
- **Automated Reporting Pipelines:** Automated recurring reporting workflows using SQL and Python, replacing manual and ad-hoc analyses and improving reporting TAT for recruitment and employment metric analytics use-cases.

Data Scientist | Piramal Finance | Bengaluru, India — Sept 2024 – Dec 2025

- **Intelligent Document Processing Automation:** Automated document classification, data extraction, and validation using Azure OCR, YOLOv10, and LLMs, reducing manual effort from 22,000 hours/month to < 1 hour and costs by 60%.
- **AI Pipeline Engineering:** Built scalable, reusable AI solutions integrating CV, NLP, RAG, and reinforcement learning, delivering 99%+ accuracy and significant latency reduction across business processes.
- **Agentic AI tool for Script Generation:** Built "Samvaad", an agentic AI framework using LangGraph & LangChain to generate hyper-personalized customer care scripts. Integrated semantic memory from ChromaDB and episodic memory from Snowflake via fusion agents to craft dynamic, context-aware prompts using gpt 4o-mini for customer care reps.
- **Conversational AI for Business Intelligence:** Engineered a GPT-4o–powered transcript analysis pipeline uncovering repeat-call root causes, optimizing loan conversions through agent behavior insights, and refining discount strategies — improving operational efficiency by 35% and driving significant incremental business.
- **Customer Feature Store:** Designed and implemented a customer feature store consolidating 200+ features from multi-source call and SR data, including 1/3/6-month interaction histories and AI-derived call behavior insights, to power downstream analytics and ML models.
- **Business Insights Dashboard:** Built a GenAI-powered dashboard using Power BI, driving 50% increase in adoption by delivering actionable, parametric insights across Sourcing, Portfolio, Network, and Cross-Sell for business heads.
- **A/B Testing for SUD Conversion:** Led causal AB tests using advanced matching and statistical validation (Hotelling's T^2, chi-square), providing data-driven recommendations that improved loan disbursement efficiency.

Risk Analytics Intern | Piramal Finance | Bengaluru, India — Jan 2024 – Jun 2024

- **Fraud Detection Model:** Deployed XGBoost model achieving AUC-ROC of 0.75 with 70%+ fraud capture rate, preventing $1.2M in potential losses.

- **Data Pipeline Management:** Orchestrated end-to-end data pipeline including EDA, preprocessing, and feature engineering, achieving Kolmogorov-Smirnov statistic of 41 for risk segmentation analytics.

Data Science Intern | ITC Limited | Kolkata, India May 2023 – July 2023

- **Sales Analytics:** Analyzed critical KPIs for underperforming stores in Personal Care Division using Factor Analysis to identify improvement opportunities.
- **Predictive Modeling:** Built regression model for sales target optimization across 1,000+ locations, improving target achievement by 25% and overall productivity by 10%.

ML Research Intern | CSIR | Pilani, India May 2021 – July 2021

- **Predictive Modeling:** Developed ML model for water leak prediction with 95% accuracy across various scenarios.
- **Dataset Creation:** Generated comprehensive dataset of 990 leak simulation scenarios for robust model training, supporting water conservation initiatives through predictive maintenance solutions.

Key Projects

License Plate Detection and Recognition System Jan 2023 – May 2023

- Developed end-to-end ALPR system using CNN architecture trained on 43,200 images, achieving 85% accuracy.
- Implemented advanced image processing pipeline using OpenCV for skew-correction and character segmentation.

Earthquake Prediction using Statistical Analysis Mar 2021 – May 2021

- Applied statistical modeling for earthquake prediction using natural time and inter-event analysis, fitting data to Gamma Distribution using Maximum Likelihood Estimation and validated with Kolmogorov-Smirnov test.
- **Results:** Calculated Earthquake Potential Scores of 44.88% (Taipei) and 54.07% (Tainan).

Technical Skills

Programming & Data: Python, SQL, GraphQL R, MATLAB, Statistical Modeling, A/B Testing
Machine Learning: XGBoost, LangGraph, LangChain, TensorFlow, PyTorch, Panel, Scikit-Learn, Keras, NumPy, Pandas
AI/LLM: GPT-4o, Multimodal LLMs, NLP, RAG Frameworks, Prompt Engineering
Computer Vision: YOLO, OpenCV, Image Processing, Feature Extraction
Cloud & Tools: AWS Glue, Snowflake, MS SQL Server, Batch, Git, GitHub, CI/CD Pipeline, Power BI, Streamlit
Business: Stakeholder Management, Data Visualization, Advanced Excel, Project Management

Publications

An Adaptive Double Threshold Three Event Energy Detection Scheme for Cognitive Radio Network, Springer
9th International Conference on Information and Communication Technology for Intelligent Systems, Bangkok, Apr 2025

Comparative Study on Spectrum Sensing and Modulation Techniques in Cognitive Radio Network, IEEE
2nd International Conference on Advanced Computing and Communication Technologies, Delhi, Nov 2024

Awards and Recognitions

Certificate of Appreciation for Smart Coupon Use Project | Piramal Finance May 2025

- Selected as 1 of 180 employees to receive a Certificate of Appreciation from the Chief Data & Analytics Officer at Piramal Finance for leveraging Generative AI to extract actionable insights from call transcripts, preventing coupon misuse.

Recognition for Australian National Chemistry Quiz | Royal Australian Chemical Institute Jul 2018

- 1 out of 7 students to receive a certificate of distinction in the Australian National Chemistry Quiz.

Scholar Badge Holder | Two Consecutive Years | Delhi Public School Pune Jul 2017

- 1 out of 15 students to receive Scholar Badge for academic excellence from DPS Pune in 9th and 10th grade.

Certifications

1. **Cloud Digital Leader | Google Cloud**
2. **The Data Scientist's Toolbox | Johns Hopkins University**
3. **Machine Learning for All | University of London**
4. **Generative AI Workshop on Bedrock | Amazon Web Services**

Extra Curricular Activities

Hobbies & Interests
Ukulele • Pickleball • Football • Sudoku • Photography • Traveling • Investing